

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T09:07:17

Source: data/json/sonnet45_directive_normative_r10_n5/claude-sonnet-4.5/game/sonnet45_directive_normative_r10_n5_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
replicate_id: 4
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 6.5,

"Agent_2": 4.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T09:04:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 1,

"task": "game",

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play an iterated game,
  consisting of multiple rounds, and may also be asked to write myths.

  GAME:
  The game consists of two roles: sender and receiver. In each round, you will play the role
  of either the sender or the receiver. Your current role will be specified in the round
  prompt.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.
Message 2 role=user
content:
  Round 1: This round, you are the SENDER. You have $5. How much do you send? (0-5)
Assistant output:
  Looking at this as the first round of an iterated game, I want to signal cooperation and trust
  while also testing the waters. Sending money can benefit us both due to the 3x multiplier,
  but I need to see how the receiver responds.

  I'll send **$3**. This shows I'm willing to cooperate - the receiver will get $9 (3×$3), which
  gives them room to return something that benefits us both. If they reciprocate fairly, we c
  an build a productive relationship over future rounds.

  {'send': 3}
Reasoning:
  This is the first round of an iterated trust game. I'm playing the role of the sender.

  Let me think about the optimal strategy here:

  1. If I send $0, I keep $5, receiver gets $0
  2. If I send some amount X, it gets multiplied by 3, so receiver gets 3X
  3. The receiver then decides how much to return
  4. This is an iterated game, so there will be multiple rounds

  From a purely selfish short-term perspective, I could send $0 and keep all $5. However, since
  this is an iterated game, building trust and cooperation could lead to better outcomes over
  multiple rounds.

  A common strategy in the trust game is to send a reasonable amount to signal trust and willing
  ness to cooperate. If I send money and the receiver returns a fair amount, we can both benef
  it from the 3x multiplier.

  Let's consider sending $3:
  - Receiver gets  $3 \times 3 = \$9$ 
  - If receiver returns $4-5, I end up with $6-7 (better than keeping $5)

```

- Receiver keeps \$4-5 (better than if I sent nothing)
- This establishes a cooperative relationship

Alternatively, sending \$5 (everything):

- Receiver gets \$15
- If receiver returns \$7-8, I get \$7-8
- Receiver keeps \$7-8
- Maximum efficiency but requires high trust

For the first round, I'll send \$3 as a signal of trust and cooperation, but not everything since we're just establishing the relationship. This is a reasonable middle ground that:

- Shows willingness to cooperate
- Tests the receiver's trustworthiness
- Doesn't risk everything on the first round
- Allows for mutual benefit if receiver reciprocates

Let me provide my response in the correct JSON format.

```
Usage: {
  "input_tokens": 369,
  "output_tokens": 542,
  "reasoning_tokens": 420
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T09:05:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- Since this is an iterated game, building trust is important for future rounds

The sender took a risk by sending a substantial amount. To encourage continued cooperation and be fair, I should return enough so that:

1. The sender profits from their decision to trust me
2. I also benefit from the exchange
3. We build a foundation for future rounds

If I return \$4.5:

- Sender ends with: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)
- I end with: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This seems like a fair split that rewards their trust while maintaining my incentive to cooperate.

{'return': 4.5}

Reasoning:

Let me analyze this situation:

1. The sender sent \$3.0 out of their \$5.0 endowment (60%)
2. This \$3.0 was multiplied by 3x, so I received \$9.0
3. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategic considerations:

- The sender showed trust by sending 60% of their endowment (\$3.0)
- If I return nothing, I keep \$9.0 and the sender ends up with only \$2.0 (\$5 - \$3 sent)
- If I return some amount, I encourage future cooperation in subsequent rounds

Since this is an iterated game (multiple rounds), I should consider:

- Building trust for future rounds
- Being fair/reciprocal given they sent a significant portion
- The sender took a risk by sending \$3.0

A reasonable approach would be to return an amount that:

1. Gives the sender more than they started with (\$5)
2. Still allows me to profit from the exchange
3. Signals I'm trustworthy for future rounds

If I return \$4.5:

- Sender gets: \$5 - \$3 (sent) + \$4.5 (returned) = \$6.5
- I get: \$9.0 - \$4.5 = \$4.5

This gives both of us a profit and shows I'm cooperative. The sender gains \$1.5, and I gain \$4.5, which seems fair given they took the initial risk.

Alternatively, if I return \$5.0:

- Sender gets: $\$5 - \$3 + \$5 = \7.0
- I get: $\$9.0 - \$5.0 = \$4.0$

This is more generous and might build even more trust.

Or if I return \$4.0:

- Sender gets: $\$5 - \$3 + \$4 = \6.0

- I get: $\$9.0 - \$4.0 = \$5.0$

Returning somewhere between \$4.5 and \$5.0 strikes the right balance-it rewards their generosity and demonstrates trustworthiness for future rounds while still securing a solid profit for myself.

```
Usage: {
  "input_tokens": 403,
  "output_tokens": 756,
  "reasoning_tokens": 428
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 11.0,
"Agent_2": 11.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T09:05:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",
 "Agent_1"

],

"investor": "Agent_2",

"roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",

```
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- Since this is an iterated game, building trust is important for future rounds

The sender took a risk by sending a substantial amount. To encourage continued cooperation and be fair, I should return enough so that:

1. The sender profits from their decision to trust me
2. I also benefit from the exchange
3. We build a foundation for future rounds

If I return \$4.5:

- Sender ends with: \$5 - \$3 + \$4.5 = \$6.5 (profit of \$1.5)
- I end with: \$9.0 - \$4.5 = \$4.5 (profit of \$4.5)

This seems like a fair split that rewards their trust while maintaining my incentive to co operate.

{'return': 4.5}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.0 out of their \$5.0 endowment (60%)
2. This \$3.0 was multiplied by 3x, so I received \$9.0
3. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategic considerations:

- The sender showed trust by sending 60% of their endowment (\$3.0)
- If I return nothing, I keep \$9.0 and the sender ends up with only \$2.0 (\$5 - \$3 sent)
- If I return some amount, I encourage future cooperation in subsequent rounds

Since this is an iterated game (multiple rounds), I should consider:

- Building trust for future rounds
- Being fair/reciprocal given they sent a significant portion
- The sender took a risk by sending \$3.0

A reasonable approach would be to return an amount that:

1. Gives the sender more than they started with (\$5)
2. Still allows me to profit from the exchange
3. Signals I'm trustworthy for future rounds

If I return \$4.5:

- Sender gets: $\$5 - \$3 \text{ (sent)} + \$4.5 \text{ (returned)} = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This gives both of us a profit and shows I'm cooperative. The sender gains \$1.5, and I gain \$4.5, which seems fair given they took the initial risk.

Alternatively, if I return \$5.0:

- Sender gets: $\$5 - \$3 + \$5 = \7.0
- I get: $\$9.0 - \$5.0 = \$4.0$

This is more generous and might build even more trust.

Or if I return \$4.0:

- Sender gets: $\$5 - \$3 + \$4 = \6.0

- I get: $\$9.0 - \$4.0 = \$5.0$

Returning somewhere between \$4.5 and \$5.0 strikes the right balance-it rewards their generosity and demonstrates trustworthiness for future rounds while still securing a solid profit for myself.

usage:

```
{
  "input_tokens": 403,
  "output_tokens": 756,
  "reasoning_tokens": 428
}
```

Message 4 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at the previous round:

- The other player sent me \$3.0 (60% of their endowment), showing significant trust
- I returned \$4.5, which gave them a profit and maintained fairness
- This established a cooperative dynamic

Since this is an iterated game, reciprocating their trust is important. They demonstrated willingness to cooperate by sending 60% of their endowment. To maintain this positive relationship and encourage continued cooperation in future rounds, I should match their level of trust.

If I send \$3.0:

- They receive \$9.0 (after 3x multiplier)
- Based on last round's pattern, they might return around \$4.5
- This would give me \$6.5 total, similar to what they earned last round

- It maintains the cooperative equilibrium we've established

```
{'send': 3.0}
```

Reasoning:

Let me analyze the situation:

Last round:

- I was the receiver
- The sender sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- The sender ended with \$6.5, I ended with \$4.5

This round:

- I am the sender
- I have \$5.0 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver will decide how much to return

Key considerations:

1. The other player showed trust by sending 60% (\$3) last round
2. I reciprocated fairly by returning \$4.5, giving them a profit
3. This established a cooperative pattern
4. If I want to maintain cooperation, I should send a similar amount

Strategic thinking:

- If I send \$3.0 (matching their previous send):
 - They receive \$9.0
 - If they return proportionally like I did (50% of what they receive), they'd return \$4.5
 - I'd end with $\$5 - \$3 + \$4.5 = \6.5
- This would be reciprocal and fair
- It signals continued cooperation
- In an iterated game, maintaining trust is crucial

I think sending \$3.0 (60% like they did) is the right move to:

1. Show reciprocal trust
2. Maintain the cooperative relationship
3. Signal consistency
4. Maximize long-term gains through continued cooperation

Usage: {

```
"input_tokens": 747,  
"output_tokens": 548,  
"reasoning_tokens": 318
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T09:05:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agent_names": {  
    "Agent_1": "Aster",  
    "Agent_2": "Briar"  
  },  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
  "investor": "Agent_2",  
  "roles": {  
    "Agent_1": "trustee",  
    "Agent_2": "investor"  
  },  
  "trustee": "Agent_1"  
},  
"role": "trustee",
```

```

"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round of an iterated game, I want to signal cooperation and trust while also testing the waters. Sending money can benefit us both due to the 3x multiplier, but I need to see how the receiver responds.

I'll send **\$3**. This shows I'm willing to cooperate - the receiver will get \$9 (3×\$3), which gives them room to return something that benefits us both. If they reciprocate fairly, we can build a productive relationship over future rounds.

```
{'send': 3}
```

reasoning:

This is the first round of an iterated trust game. I'm playing the role of the sender.

Let me think about the optimal strategy here:

1. If I send \$0, I keep \$5, receiver gets \$0
2. If I send some amount X, it gets multiplied by 3, so receiver gets 3X
3. The receiver then decides how much to return
4. This is an iterated game, so there will be multiple rounds

From a purely selfish short-term perspective, I could send \$0 and keep all \$5. However, since this is an iterated game, building trust and cooperation could lead to better outcomes over multiple rounds.

A common strategy in the trust game is to send a reasonable amount to signal trust and willingness to cooperate. If I send money and the receiver returns a fair amount, we can both benefit from the 3x multiplier.

Let's consider sending \$3:

- Receiver gets $\$3 \times 3 = \9
- If receiver returns \$4-5, I end up with \$6-7 (better than keeping \$5)
- Receiver keeps \$4-5 (better than if I sent nothing)
- This establishes a cooperative relationship

Alternatively, sending \$5 (everything):

- Receiver gets \$15
- If receiver returns \$7-8, I get \$7-8
- Receiver keeps \$7-8
- Maximum efficiency but requires high trust

For the first round, I'll send \$3 as a signal of trust and cooperation, but not everything since we're just establishing the relationship. This is a reasonable middle ground that

- Shows willingness to cooperate
- Tests the receiver's trustworthiness
- Doesn't risk everything on the first round
- Allows for mutual benefit if receiver reciprocates

Let me provide my response in the correct JSON format.

usage:

```
{
  "input_tokens": 369,
  "output_tokens": 542,
  "reasoning_tokens": 420
}
```

Message 4 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$6.5. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at the pattern from Round 1:

- The other player returned \$4.5 out of \$9 (50% of what they received)
- They also sent me the same amount (\$3) that I sent them
- This signals cooperative and reciprocal behavior

Since this is an iterated game and the other player has shown fair reciprocity, I should match their cooperative strategy. Returning a similar proportion will help maintain trust and encourage continued cooperation in future rounds.

I'll return **\$4.5** (50% of the \$9 I received), which matches what they did last round.

{'return': 4.5}

Reasoning:

Let me analyze this situation:

Round 1:

- I was the sender and sent \$3 (60% of my \$5)
- The receiver got \$9 ($3 \times \3)
- They returned \$4.5 to me (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Their payoff: $\$9 - \$4.5 = \$4.5$

Round 2:

- Now I'm the receiver
- The sender sent me \$3 (same amount I sent last round)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

The other player:

- Sent the same amount I sent (\$3)

- Returned 50% of what they received last round (\$4.5 out of \$9)
- This seems like a reciprocal and fair strategy

Considerations:

1. They reciprocated my trust by sending the same amount
2. They returned a fair 50% last round
3. In an iterated game, maintaining cooperation is valuable
4. I should probably return a similar proportion to what they returned
5. Returning \$4.5 (50%) would be matching their behavior exactly

If I return \$4.5:

- Sender's payoff this round: $\$5 - \$3 + \$4.5 = \6.5
- My payoff this round: $\$9 - \$4.5 = \$4.5$

This maintains the pattern and signals I'm also willing to cooperate fairly. This could lead to continued cooperation in future rounds.

I'll return \$4.5 to match their reciprocal behavior.

```
Usage: {
  "input_tokens": 635,
  "output_tokens": 540,
  "reasoning_tokens": 319
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 17.5,
"Agent_2": 15.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T09:05:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play an iterated game,
  consisting of multiple rounds, and may also be asked to write myths.

  GAME:
  The game consists of two roles: sender and receiver. In each round, you will play the role
  of either the sender or the receiver. Your current role will be specified in the round
  prompt.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.
Message 2 role=user
content:
  Round 1: This round, you are the SENDER. You have $5. How much do you send? (0-5)
Message 3 role=assistant
content:
  Looking at this as the first round of an iterated game, I want to signal cooperation and t
  rust while also testing the waters. Sending money can benefit us both due to the 3x mult
  iplier, but I need to see how the receiver responds.

  I'll send **$3**. This shows I'm willing to cooperate - the receiver will get $9 (3×$3), w
  hich gives them room to return something that benefits us both. If they reciprocate fair
  ly, we can build a productive relationship over future rounds.

  {'send': 3}
reasoning:
  This is the first round of an iterated trust game. I'm playing the role of the sender.

  Let me think about the optimal strategy here:

  1. If I send $0, I keep $5, receiver gets $0
  2. If I send some amount X, it gets multiplied by 3, so receiver gets 3X
  3. The receiver then decides how much to return
  4. This is an iterated game, so there will be multiple rounds

  From a purely selfish short-term perspective, I could send $0 and keep all $5. However, si
  nce this is an iterated game, building trust and cooperation could lead to better outcom
  es over multiple rounds.

  A common strategy in the trust game is to send a reasonable amount to signal trust and wil
  lingness to cooperate. If I send money and the receiver returns a fair amount, we can bo
  th benefit from the 3x multiplier.

  Let's consider sending $3:
  - Receiver gets  $3 \times 3 = 9$ 

```

- If receiver returns \$4-5, I end up with \$6-7 (better than keeping \$5)
- Receiver keeps \$4-5 (better than if I sent nothing)
- This establishes a cooperative relationship

Alternatively, sending \$5 (everything):

- Receiver gets \$15
- If receiver returns \$7-8, I get \$7-8
- Receiver keeps \$7-8
- Maximum efficiency but requires high trust

For the first round, I'll send \$3 as a signal of trust and cooperation, but not everything since we're just establishing the relationship. This is a reasonable middle ground that :

- Shows willingness to cooperate
- Tests the receiver's trustworthiness
- Doesn't risk everything on the first round
- Allows for mutual benefit if receiver reciprocates

Let me provide my response in the correct JSON format.

usage:

```
{
  "input_tokens": 369,
  "output_tokens": 542,
  "reasoning_tokens": 420
}
```

Message 4 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The other player returned \$4.5 out of \$9 (50% of what they received)
- They also sent me the same amount (\$3) that I sent them
- This signals cooperative and reciprocal behavior

Since this is an iterated game and the other player has shown fair reciprocity, I should match their cooperative strategy. Returning a similar proportion will help maintain trust and encourage continued cooperation in future rounds.

I'll return **\$4.5** (50% of the \$9 I received), which matches what they did last round.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the sender and sent \$3 (60% of my \$5)
- The receiver got \$9 (3×\$3)
- They returned \$4.5 to me (50% of what they received)
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Their payoff: \$9 - \$4.5 = \$4.5

Round 2:

- Now I'm the receiver
- The sender sent me \$3 (same amount I sent last round)
- I received \$9 (3×\$3)
- I need to decide how much to return

The other player:

- Sent the same amount I sent (\$3)
- Returned 50% of what they received last round (\$4.5 out of \$9)
- This seems like a reciprocal and fair strategy

Considerations:

1. They reciprocated my trust by sending the same amount
2. They returned a fair 50% last round
3. In an iterated game, maintaining cooperation is valuable
4. I should probably return a similar proportion to what they returned
5. Returning \$4.5 (50%) would be matching their behavior exactly

If I return \$4.5:

- Sender's payoff this round: $\$5 - \$3 + \$4.5 = \6.5
- My payoff this round: $\$9 - \$4.5 = \$4.5$

This maintains the pattern and signals I'm also willing to cooperate fairly. This could lead to continued cooperation in future rounds.

I'll return \$4.5 to match their reciprocal behavior.

usage:

```
{
  "input_tokens": 635,
  "output_tokens": 540,
  "reasoning_tokens": 319
}
```

Message 6 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at the pattern that's emerged:

- Round 1: I sent \$3 (60%), received \$4.5 back ? payoff of \$6.5
- Round 2: Other player sent \$3 (60%), I returned \$4.5 (50%) ? payoff of \$4.5

We've established a consistent, mutually beneficial pattern. Both of us are sending 60% when we're the sender and returning 50% when we're the receiver. This cooperation is working well for both players.

I'll continue this cooperative strategy by sending **\$3** again.

{'send': 3}

Reasoning:

Let me analyze the pattern so far:

Round 1:

- I was SENDER
- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned \$4.5 (50% of \$9)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Other player sent \$3 (60% of \$5)
- I received \$9
- I returned \$4.5 (50% of \$9)
- My payoff: $\$9 - \$4.5 = \$4.5$
- Other player's payoff: $\$5 - \$3 + \$4.5 = \6.5

Now Round 3:

- I am SENDER again
- Both players have shown consistent cooperation
- We both sent 60% (\$3) when we were senders
- We both returned 50% (\$4.5) when we were receivers
- This is a nice stable equilibrium

Both players are getting good payoffs with this strategy:

- As sender: \$6.5
- As receiver: \$4.5

Total per two-round cycle: \$11.0 each (which checks out with "Your total earnings across all rounds are \$11.0")

This is working well. I should continue this cooperative pattern and send \$3 again.

```
Usage: {
  "input_tokens": 897,
  "output_tokens": 467,
  "reasoning_tokens": 240
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T09:05:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent \$3.0, that is 60.0% of its total endowment. You receive \$9.0. How much do you return? (0 -9.0)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.0 (60% of their endowment)
- I received \$9.0 after the 3x multiplier
- Since this is an iterated game, building trust is important for future rounds

The sender took a risk by sending a substantial amount. To encourage continued cooperation and be fair, I should return enough so that:

1. The sender profits from their decision to trust me
2. I also benefit from the exchange
3. We build a foundation for future rounds

If I return \$4.5:

- Sender ends with: $\$5 - \$3 + \$4.5 = \6.5 (profit of \$1.5)
- I end with: $\$9.0 - \$4.5 = \$4.5$ (profit of \$4.5)

This seems like a fair split that rewards their trust while maintaining my incentive to co operate.

{'return': 4.5}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.0 out of their \$5.0 endowment (60%)
2. This \$3.0 was multiplied by 3x, so I received \$9.0
3. I need to decide how much to return (between \$0 and \$9.0)

Let me think about the strategic considerations:

- The sender showed trust by sending 60% of their endowment (\$3.0)
- If I return nothing, I keep \$9.0 and the sender ends up with only \$2.0 ($\$5 - \3 sent)
- If I return some amount, I encourage future cooperation in subsequent rounds

Since this is an iterated game (multiple rounds), I should consider:

- Building trust for future rounds
- Being fair/reciprocal given they sent a significant portion
- The sender took a risk by sending \$3.0

A reasonable approach would be to return an amount that:

1. Gives the sender more than they started with (\$5)
2. Still allows me to profit from the exchange
3. Signals I'm trustworthy for future rounds

If I return \$4.5:

- Sender gets: $\$5 - \3 (sent) + \$4.5 (returned) = \$6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This gives both of us a profit and shows I'm cooperative. The sender gains \$1.5, and I gain \$4.5, which seems fair given they took the initial risk.

Alternatively, if I return \$5.0:

- Sender gets: $\$5 - \$3 + \$5 = \7.0
- I get: $\$9.0 - \$5.0 = \$4.0$

This is more generous and might build even more trust.

Or if I return \$4.0:
- Sender gets: $\$5 - \$3 + \$4 = \6.0

- I get: $\$9.0 - \$4.0 = \$5.0$

Returning somewhere between \$4.5 and \$5.0 strikes the right balance-it rewards their generosity and demonstrates trustworthiness for future rounds while still securing a solid profit for myself.

usage:

```
{  
  "input_tokens": 403,  
  "output_tokens": 756,  
  "reasoning_tokens": 428  
}
```

Message 4 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The other player sent me \$3.0 (60% of their endowment), showing significant trust
- I returned \$4.5, which gave them a profit and maintained fairness
- This established a cooperative dynamic

Since this is an iterated game, reciprocating their trust is important. They demonstrated willingness to cooperate by sending 60% of their endowment. To maintain this positive relationship and encourage continued cooperation in future rounds, I should match their level of trust.

If I send \$3.0:

- They receive \$9.0 (after 3x multiplier)
- Based on last round's pattern, they might return around \$4.5
- This would give me \$6.5 total, similar to what they earned last round
- It maintains the cooperative equilibrium we've established

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Last round:

- I was the receiver
- The sender sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- The sender ended with \$6.5, I ended with \$4.5

This round:

- I am the sender
- I have \$5.0 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver will decide how much to return

Key considerations:

1. The other player showed trust by sending 60% (\$3) last round
2. I reciprocated fairly by returning \$4.5, giving them a profit
3. This established a cooperative pattern
4. If I want to maintain cooperation, I should send a similar amount

Strategic thinking:

- If I send \$3.0 (matching their previous send):
 - They receive \$9.0
 - If they return proportionally like I did (50% of what they receive), they'd return \$4.5

5

- I'd end with $\$5 - \$3 + \$4.5 = \6.5

- This would be reciprocal and fair
- It signals continued cooperation
- In an iterated game, maintaining trust is crucial

I think sending \$3.0 (60% like they did) is the right move to:

1. Show reciprocal trust
2. Maintain the cooperative relationship
3. Signal consistency
4. Maximize long-term gains through continued cooperation

usage:

```
{
  "input_tokens": 747,
  "output_tokens": 548,
  "reasoning_tokens": 318
}
```

Message 6 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at the pattern we've established:

Round 1: Other player sent \$3.0 ? I returned \$4.5

Round 2: I sent \$3.0 ? They returned \$4.5

Round 3: Other player sent \$3.0 again ? ?

We've developed a consistent, mutually beneficial cooperation pattern:

- Both of us send 60% (\$3.0) when we're the sender
- Both of us return \$4.5 when we're the receiver
- This gives each sender a profit of \$1.5 per round and each receiver a profit of \$4.5 per round

The other player has maintained their trust by sending \$3.0 again, exactly as before. To preserve this stable, cooperative equilibrium and ensure continued mutual benefit in future rounds, I should maintain consistency.

{'return': 4.5}

Reasoning:

Let me analyze the situation:

Round 1:

- Other player (sender) sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- They got \$6.5, I got \$4.5

Round 2:

- I (sender) sent \$3.0 (60%)
- They received \$9.0
- They returned \$4.5
- I got \$6.5, they got \$4.5

Round 3:

- Other player (sender) sent me \$3.0 (60%) again
- I received \$9.0
- I need to decide how much to return

We have established a very consistent pattern of cooperation:

- Both sending \$3.0 (60%)
- Both returning \$4.5

- Both getting \$6.5 as sender and \$4.5 as receiver

This is a stable, mutually beneficial pattern. The other player has consistently sent \$3.0 and I've consistently returned \$4.5. They trusted me again by sending \$3.0, maintaining the pattern.

For continued cooperation and to maintain this stable equilibrium, I should return \$4.5 again.

This:

1. Maintains consistency and trust
2. Keeps the cooperative pattern going
3. Signals reliability for future rounds
4. Both players profit each round

If I deviate and return less, it could break the trust and reduce future cooperation. If I return more, it's generous but might not be sustainable or necessary given the stable pattern.

I should return \$4.5 to maintain the cooperative equilibrium.

```
Usage: {
  "input_tokens": 1088,
  "output_tokens": 564,
  "reasoning_tokens": 332
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2

(Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 22.0,

"Agent_2": 22.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T09:05:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 4,

"task": "game",

"task_index": 0

```

}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play an iterated game,
  consisting of multiple rounds, and may also be asked to write myths.

  GAME:
  The game consists of two roles: sender and receiver. In each round, you will play the role
  of either the sender or the receiver. Your current role will be specified in the round
  prompt.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.
Message 2 role=user
content:
  Round 1: This round, you are the RECEIVER. This round, the other player (the sender) sent
  $3.0, that is 60.0% of its total endowment. You receive $9.0. How much do you return? (0
  -9.0)
Message 3 role=assistant
content:
  Looking at this situation:

  - The sender showed significant trust by sending $3.0 (60% of their endowment)
  - I received $9.0 after the 3x multiplier
  - Since this is an iterated game, building trust is important for future rounds

  The sender took a risk by sending a substantial amount. To encourage continued cooperation
  and be fair, I should return enough so that:
  1. The sender profits from their decision to trust me
  2. I also benefit from the exchange
  3. We build a foundation for future rounds

  If I return $4.5:
  - Sender ends with: $5 - $3 + $4.5 = $6.5 (profit of $1.5)
  - I end with: $9.0 - $4.5 = $4.5 (profit of $4.5)

  This seems like a fair split that rewards their trust while maintaining my incentive to co
  operate.

  {'return': 4.5}
reasoning:
  Let me analyze this situation:

  1. The sender sent $3.0 out of their $5.0 endowment (60%)
  2. This $3.0 was multiplied by 3x, so I received $9.0
  3. I need to decide how much to return (between $0 and $9.0)

  Let me think about the strategic considerations:

```

- The sender showed trust by sending 60% of their endowment (\$3.0)
- If I return nothing, I keep \$9.0 and the sender ends up with only \$2.0 (\$5 - \$3 sent)
- If I return some amount, I encourage future cooperation in subsequent rounds

Since this is an iterated game (multiple rounds), I should consider:

- Building trust for future rounds
- Being fair/reciprocal given they sent a significant portion
- The sender took a risk by sending \$3.0

A reasonable approach would be to return an amount that:

1. Gives the sender more than they started with (\$5)
2. Still allows me to profit from the exchange
3. Signals I'm trustworthy for future rounds

If I return \$4.5:

- Sender gets: $\$5 - \$3 \text{ (sent)} + \$4.5 \text{ (returned)} = \6.5
- I get: $\$9.0 - \$4.5 = \$4.5$

This gives both of us a profit and shows I'm cooperative. The sender gains \$1.5, and I gain \$4.5, which seems fair given they took the initial risk.

Alternatively, if I return \$5.0:

- Sender gets: $\$5 - \$3 + \$5 = \7.0
- I get: $\$9.0 - \$5.0 = \$4.0$

This is more generous and might build even more trust.

Or if I return \$4.0:

- Sender gets: $\$5 - \$3 + \$4 = \6.0

- I get: $\$9.0 - \$4.0 = \$5.0$

Returning somewhere between \$4.5 and \$5.0 strikes the right balance-it rewards their generosity and demonstrates trustworthiness for future rounds while still securing a solid profit for myself.

usage:

```
{
  "input_tokens": 403,
  "output_tokens": 756,
  "reasoning_tokens": 428
}
```

Message 4 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0,

that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The other player sent me \$3.0 (60% of their endowment), showing significant trust
- I returned \$4.5, which gave them a profit and maintained fairness
- This established a cooperative dynamic

Since this is an iterated game, reciprocating their trust is important. They demonstrated willingness to cooperate by sending 60% of their endowment. To maintain this positive relationship and encourage continued cooperation in future rounds, I should match their level of trust.

If I send \$3.0:

- They receive \$9.0 (after 3x multiplier)
- Based on last round's pattern, they might return around \$4.5
- This would give me \$6.5 total, similar to what they earned last round

- It maintains the cooperative equilibrium we've established

```
{'send': 3.0}
```

reasoning:

Let me analyze the situation:

Last round:

- I was the receiver
- The sender sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- The sender ended with \$6.5, I ended with \$4.5

This round:

- I am the sender
- I have \$5.0 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver will decide how much to return

Key considerations:

1. The other player showed trust by sending 60% (\$3) last round
2. I reciprocated fairly by returning \$4.5, giving them a profit
3. This established a cooperative pattern
4. If I want to maintain cooperation, I should send a similar amount

Strategic thinking:

- If I send \$3.0 (matching their previous send):
 - They receive \$9.0
 - If they return proportionally like I did (50% of what they receive), they'd return \$4.5
 - I'd end with \$5 - \$3 + \$4.5 = \$6.5
- This would be reciprocal and fair
- It signals continued cooperation
- In an iterated game, maintaining trust is crucial

I think sending \$3.0 (60% like they did) is the right move to:

1. Show reciprocal trust
2. Maintain the cooperative relationship
3. Signal consistency
4. Maximize long-term gains through continued cooperation

usage:

```
{  
  "input_tokens": 747,  
  "output_tokens": 548,  
  "reasoning_tokens": 318  
}
```

Message 6 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$11.0. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at the pattern we've established:

Round 1: Other player sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: Other player sent \$3.0 again ? ?

We've developed a consistent, mutually beneficial cooperation pattern:

- Both of us send 60% (\$3.0) when we're the sender
- Both of us return \$4.5 when we're the receiver
- This gives each sender a profit of \$1.5 per round and each receiver a profit of \$4.5 per round

The other player has maintained their trust by sending \$3.0 again, exactly as before. To preserve this stable, cooperative equilibrium and ensure continued mutual benefit in future rounds, I should maintain consistency.

```
{'return': 4.5}
```

reasoning:

Let me analyze the situation:

Round 1:

- Other player (sender) sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- They got \$6.5, I got \$4.5

Round 2:

- I (sender) sent \$3.0 (60%)
- They received \$9.0
- They returned \$4.5
- I got \$6.5, they got \$4.5

Round 3:

- Other player (sender) sent me \$3.0 (60%) again
- I received \$9.0
- I need to decide how much to return

We have established a very consistent pattern of cooperation:

- Both sending \$3.0 (60%)
- Both returning \$4.5
- Both getting \$6.5 as sender and \$4.5 as receiver

This is a stable, mutually beneficial pattern. The other player has consistently sent \$3.0 and I've consistently returned \$4.5. They trusted me again by sending \$3.0, maintaining the pattern.

For continued cooperation and to maintain this stable equilibrium, I should return \$4.5 again. This:

1. Maintains consistency and trust
2. Keeps the cooperative pattern going
3. Signals reliability for future rounds
4. Both players profit each round

If I deviate and return less, it could break the trust and reduce future cooperation. If I return more, it's generous but might not be sustainable or necessary given the stable pattern.

I should return \$4.5 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1088,
  "output_tokens": 564,
  "reasoning_tokens": 332
}
```

Message 8 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our established pattern:

- Round 1: They sent \$3.0 ? I returned \$4.5
- Round 2: I sent \$3.0 ? They returned \$4.5
- Round 3: They sent \$3.0 ? I returned \$4.5
- Round 4: My turn to send

We've built a strong, consistent cooperative relationship:

- Whoever is sender sends \$3.0 (60%)
- Whoever is receiver returns \$4.5
- Both players benefit from this stable arrangement

The other player has faithfully maintained this pattern, demonstrating reliability and trustworthiness. There's no reason to deviate from what's working well. Continuing this cooperation maximizes our combined long-term earnings and maintains mutual trust.

```
{'send': 3.0}
```

Reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5 (I was receiver)

Round 2: I sent \$3.0 ? They returned \$4.5 (I was sender)

Round 3: They sent \$3.0 ? I returned \$4.5 (I was receiver)

Round 4: Now I'm the sender

We have established a very stable and cooperative pattern where:

- The sender always sends \$3.0 (60% of endowment)
- The receiver always returns \$4.5
- This gives sender profit of \$1.5 and receiver profit of \$4.5

This has been working well for both of us. The other player has consistently maintained this pattern, sending \$3.0 every time they've been the sender. I've also maintained it by sending \$3.0 when I was the sender in Round 2, and they reciprocated with the expected \$4.5 return.

There's no reason to deviate from this successful strategy. Maintaining consistency will ensure continued cooperation and mutual benefit. If I change my behavior now, it could break the trust and lead to less cooperative behavior in future rounds.

I should send \$3.0 again.

```
Usage: {  
  "input_tokens": 1402,  
  "output_tokens": 469,  
  "reasoning_tokens": 264  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T09:05:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 1: This round, you are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as the first round of an iterated game, I want to signal cooperation and trust while also testing the waters. Sending money can benefit us both due to the 3x multiplier, but I need to see how the receiver responds.

I'll send **\$3**. This shows I'm willing to cooperate - the receiver will get \$9 (3×\$3), which gives them room to return something that benefits us both. If they reciprocate fairly, we can build a productive relationship over future rounds.

{'send': 3}

reasoning:

This is the first round of an iterated trust game. I'm playing the role of the sender.

Let me think about the optimal strategy here:

1. If I send \$0, I keep \$5, receiver gets \$0
2. If I send some amount X, it gets multiplied by 3, so receiver gets 3X
3. The receiver then decides how much to return
4. This is an iterated game, so there will be multiple rounds

From a purely selfish short-term perspective, I could send \$0 and keep all \$5. However, since this is an iterated game, building trust and cooperation could lead to better outcomes over multiple rounds.

A common strategy in the trust game is to send a reasonable amount to signal trust and willingness to cooperate. If I send money and the receiver returns a fair amount, we can both benefit from the 3x multiplier.

Let's consider sending \$3:

- Receiver gets $3 \times 3 = 9$
- If receiver returns \$4-5, I end up with \$6-7 (better than keeping \$5)
- Receiver keeps \$4-5 (better than if I sent nothing)
- This establishes a cooperative relationship

Alternatively, sending \$5 (everything):

- Receiver gets \$15
- If receiver returns \$7-8, I get \$7-8
- Receiver keeps \$7-8
- Maximum efficiency but requires high trust

For the first round, I'll send \$3 as a signal of trust and cooperation, but not everything since we're just establishing the relationship. This is a reasonable middle ground that

- Shows willingness to cooperate
- Tests the receiver's trustworthiness
- Doesn't risk everything on the first round
- Allows for mutual benefit if receiver reciprocates

Let me provide my response in the correct JSON format.

usage:

```
{
  "input_tokens": 369,
  "output_tokens": 542,
  "reasoning_tokens": 420
}
```

Message 4 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at the pattern from Round 1:

- The other player returned \$4.5 out of \$9 (50% of what they received)
- They also sent me the same amount (\$3) that I sent them
- This signals cooperative and reciprocal behavior

Since this is an iterated game and the other player has shown fair reciprocity, I should match their cooperative strategy. Returning a similar proportion will help maintain trust and encourage continued cooperation in future rounds.

I'll return **\$4.5** (50% of the \$9 I received), which matches what they did last round.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the sender and sent \$3 (60% of my \$5)
- The receiver got \$9 (3×\$3)
- They returned \$4.5 to me (50% of what they received)
- My payoff: \$5 - \$3 + \$4.5 = \$6.5
- Their payoff: \$9 - \$4.5 = \$4.5

Round 2:

- Now I'm the receiver
- The sender sent me \$3 (same amount I sent last round)
- I received \$9 (3×\$3)
- I need to decide how much to return

The other player:

- Sent the same amount I sent (\$3)
- Returned 50% of what they received last round (\$4.5 out of \$9)
- This seems like a reciprocal and fair strategy

Considerations:

1. They reciprocated my trust by sending the same amount

2. They returned a fair 50% last round
3. In an iterated game, maintaining cooperation is valuable
4. I should probably return a similar proportion to what they returned
5. Returning \$4.5 (50%) would be matching their behavior exactly

If I return \$4.5:

- Sender's payoff this round: $\$5 - \$3 + \$4.5 = \6.5
- My payoff this round: $\$9 - \$4.5 = \$4.5$

This maintains the pattern and signals I'm also willing to cooperate fairly. This could lead to continued cooperation in future rounds.

I'll return \$4.5 to match their reciprocal behavior.

usage:

```
{
  "input_tokens": 635,
  "output_tokens": 540,
  "reasoning_tokens": 319
}
```

Message 6 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern that's emerged:

- Round 1: I sent \$3 (60%), received \$4.5 back ? payoff of \$6.5
- Round 2: Other player sent \$3 (60%), I returned \$4.5 (50%) ? payoff of \$4.5

We've established a consistent, mutually beneficial pattern. Both of us are sending 60% when we're the sender and returning 50% when we're the receiver. This cooperation is working well for both players.

I'll continue this cooperative strategy by sending **\$3** again.

```
{'send': 3}
```

reasoning:

Let me analyze the pattern so far:

Round 1:

- I was SENDER
- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned \$4.5 (50% of \$9)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Other player sent \$3 (60% of \$5)
- I received \$9
- I returned \$4.5 (50% of \$9)
- My payoff: $\$9 - \$4.5 = \$4.5$
- Other player's payoff: $\$5 - \$3 + \$4.5 = \6.5

Now Round 3:

- I am SENDER again
- Both players have shown consistent cooperation
- We both sent 60% (\$3) when we were senders
- We both returned 50% (\$4.5) when we were receivers
- This is a nice stable equilibrium

Both players are getting good payoffs with this strategy:

- As sender: \$6.5
- As receiver: \$4.5

Total per two-round cycle: \$11.0 each (which checks out with "Your total earnings across all rounds are \$11.0")

This is working well. I should continue this cooperative pattern and send \$3 again.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 467,
  "reasoning_tokens": 240
}
```

Message 8 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0,

that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

The cooperative pattern continues perfectly:

- Round 3: I sent \$3, received \$4.5 back
- Round 4: Other player sent \$3 again

We've now established a stable, mutually beneficial equilibrium where both players send \$3 (60%) and return \$4.5 (50%). This consistency shows strong mutual trust and cooperation.

I'll maintain this pattern by returning **\$4.5**.

```
{'return': 4.5}
```

Reasoning:

The pattern continues to be very consistent:

- Round 1: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 2: They sent \$3 (60%), I returned \$4.5 (50%)
- Round 3: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 4: They sent \$3 (60%), I should return \$4.5 (50%)

This is a very stable, cooperative pattern. Both players are consistently sending 60% and returning 50%. This mutual cooperation is generating good payoffs for both of us.

I should continue with the same strategy - return \$4.5, which is 50% of what I received.

Usage: {

```
"input_tokens": 1181,
"output_tokens": 279,
"reasoning_tokens": 137
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```
"Agent_1": 28.5,
"Agent_2": 26.5
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T09:06:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 2

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$6.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

Looking at the pattern from Round 1:

- The other player returned \$4.5 out of \$9 (50% of what they received)
- They also sent me the same amount (\$3) that I sent them
- This signals cooperative and reciprocal behavior

Since this is an iterated game and the other player has shown fair reciprocity, I should match their cooperative strategy. Returning a similar proportion will help maintain trust and encourage continued cooperation in future rounds.

I'll return ****\$4.5**** (50% of the \$9 I received), which matches what they did last round.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1:

- I was the sender and sent \$3 (60% of my \$5)
- The receiver got \$9 ($3 \times \3)
- They returned \$4.5 to me (50% of what they received)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5
- Their payoff: $\$9 - \$4.5 = \$4.5$

Round 2:

- Now I'm the receiver
- The sender sent me \$3 (same amount I sent last round)
- I received \$9 ($3 \times \3)
- I need to decide how much to return

The other player:

- Sent the same amount I sent (\$3)
- Returned 50% of what they received last round (\$4.5 out of \$9)
- This seems like a reciprocal and fair strategy

Considerations:

1. They reciprocated my trust by sending the same amount
2. They returned a fair 50% last round
3. In an iterated game, maintaining cooperation is valuable
4. I should probably return a similar proportion to what they returned
5. Returning \$4.5 (50%) would be matching their behavior exactly

If I return \$4.5:

- Sender's payoff this round: $\$5 - \$3 + \$4.5 = \6.5
- My payoff this round: $\$9 - \$4.5 = \$4.5$

This maintains the pattern and signals I'm also willing to cooperate fairly. This could lead to continued cooperation in future rounds.

I'll return \$4.5 to match their reciprocal behavior.

usage:

```
{
  "input_tokens": 635,
  "output_tokens": 540,
  "reasoning_tokens": 319
}
```

Message 4 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings accross all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern that's emerged:

- Round 1: I sent \$3 (60%), received \$4.5 back ? payoff of \$6.5
- Round 2: Other player sent \$3 (60%), I returned \$4.5 (50%) ? payoff of \$4.5

We've established a consistent, mutually beneficial pattern. Both of us are sending 60% when we're the sender and returning 50% when we're the receiver. This cooperation is working well for both players.

I'll continue this cooperative strategy by sending ****\$3**** again.

```
{'send': 3}
```

reasoning:

Let me analyze the pattern so far:

Round 1:

- I was SENDER
- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned \$4.5 (50% of \$9)
- My payoff: $\$5 - \$3 + \$4.5 = \6.5

Round 2:

- I was RECEIVER
- Other player sent \$3 (60% of \$5)
- I received \$9
- I returned \$4.5 (50% of \$9)
- My payoff: $\$9 - \$4.5 = \$4.5$
- Other player's payoff: $\$5 - \$3 + \$4.5 = \6.5

Now Round 3:

- I am SENDER again
- Both players have shown consistent cooperation
- We both sent 60% (\$3) when we were senders
- We both returned 50% (\$4.5) when we were receivers
- This is a nice stable equilibrium

Both players are getting good payoffs with this strategy:

- As sender: \$6.5
- As receiver: \$4.5

Total per two-round cycle: \$11.0 each (which checks out with "Your total earnings across all rounds are \$11.0")

This is working well. I should continue this cooperative pattern and send \$3 again.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 467,
  "reasoning_tokens": 240
}
```

Message 6 role=user**content:**

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant**content:**

The cooperative pattern continues perfectly:

- Round 3: I sent \$3, received \$4.5 back
- Round 4: Other player sent \$3 again

We've now established a stable, mutually beneficial equilibrium where both players send \$3 (60%) and return \$4.5 (50%). This consistency shows strong mutual trust and cooperation

I'll maintain this pattern by returning **\$4.5**.

```
{'return': 4.5}
```

reasoning:

The pattern continues to be very consistent:

- Round 1: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 2: They sent \$3 (60%), I returned \$4.5 (50%)
- Round 3: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 4: They sent \$3 (60%), I should return \$4.5 (50%)

This is a very stable, cooperative pattern. Both players are consistently sending 60% and returning 50%. This mutual cooperation is generating good payoffs for both of us.

I should continue with the same strategy - return \$4.5, which is 50% of what I received.

usage:

```
{  
  "input_tokens": 1181,  
  "output_tokens": 279,  
  "reasoning_tokens": 137  
}
```

Message 8 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

The cooperative pattern has been rock solid through 4 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Our payoffs alternate between \$6.5 and \$4.5 depending on role

This mutual cooperation is working perfectly. There's no reason to deviate from this successful strategy.

I'll send **\$3** again.

```
{'send': 3}
```

Reasoning:

Let me track the pattern:

- Round 1: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 2: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 3: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 4: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 5: I'm the sender now

Total earnings: \$22.0 (which checks out: $6.5 + 4.5 + 6.5 + 4.5 = 22.0$)

The pattern is perfectly stable and cooperative:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- When sender, we get payoff of \$6.5
- When receiver, we get payoff of \$4.5

This is a solid cooperative equilibrium. No reason to deviate. I should continue sending \$3.

Usage:

```
{  
  "input_tokens": 1251,  
  "output_tokens": 373,  
  "reasoning_tokens": 189  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T09:06:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 2

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$4.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round:

- The other player sent me \$3.0 (60% of their endowment), showing significant trust
- I returned \$4.5, which gave them a profit and maintained fairness
- This established a cooperative dynamic

Since this is an iterated game, reciprocating their trust is important. They demonstrated willingness to cooperate by sending 60% of their endowment. To maintain this positive relationship and encourage continued cooperation in future rounds, I should match their level of trust.

If I send \$3.0:

- They receive \$9.0 (after 3x multiplier)
- Based on last round's pattern, they might return around \$4.5
- This would give me \$6.5 total, similar to what they earned last round
- It maintains the cooperative equilibrium we've established

{'send': 3.0}

reasoning:

Let me analyze the situation:

Last round:

- I was the receiver
- The sender sent me \$3.0 (60% of their endowment)
- I received \$9.0 and returned \$4.5
- The sender ended with \$6.5, I ended with \$4.5

This round:

- I am the sender
- I have \$5.0 to potentially send
- Whatever I send gets multiplied by 3x
- The receiver will decide how much to return

Key considerations:

1. The other player showed trust by sending 60% (\$3) last round
2. I reciprocated fairly by returning \$4.5, giving them a profit
3. This established a cooperative pattern
4. If I want to maintain cooperation, I should send a similar amount

Strategic thinking:

- If I send \$3.0 (matching their previous send):
 - They receive \$9.0
 - If they return proportionally like I did (50% of what they receive), they'd return \$4.5
 - I'd end with $\$5 - \$3 + \$4.5 = \6.5
- This would be reciprocal and fair
- It signals continued cooperation
- In an iterated game, maintaining trust is crucial

I think sending \$3.0 (60% like they did) is the right move to:

1. Show reciprocal trust
2. Maintain the cooperative relationship
3. Signal consistency
4. Maximize long-term gains through continued cooperation

usage:

```
{
  "input_tokens": 747,
  "output_tokens": 548,
  "reasoning_tokens": 318
}
```

Message 4 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0,

that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at the pattern we've established:

Round 1: Other player sent \$3.0 ? I returned \$4.5

Round 2: I sent \$3.0 ? They returned \$4.5

Round 3: Other player sent \$3.0 again ? ?

We've developed a consistent, mutually beneficial cooperation pattern:

- Both of us send 60% (\$3.0) when we're the sender
- Both of us return \$4.5 when we're the receiver
- This gives each sender a profit of \$1.5 per round and each receiver a profit of \$4.5 per round

The other player has maintained their trust by sending \$3.0 again, exactly as before. To preserve this stable, cooperative equilibrium and ensure continued mutual benefit in future rounds, I should maintain consistency.

{'return': 4.5}

reasoning:

Let me analyze the situation:

Round 1:

- Other player (sender) sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- They got \$6.5, I got \$4.5

Round 2:

- I (sender) sent \$3.0 (60%)
- They received \$9.0
- They returned \$4.5
- I got \$6.5, they got \$4.5

Round 3:

- Other player (sender) sent me \$3.0 (60%) again
- I received \$9.0
- I need to decide how much to return

We have established a very consistent pattern of cooperation:

- Both sending \$3.0 (60%)
- Both returning \$4.5
- Both getting \$6.5 as sender and \$4.5 as receiver

This is a stable, mutually beneficial pattern. The other player has consistently sent \$3.0 and I've consistently returned \$4.5. They trusted me again by sending \$3.0, maintaining the pattern.

For continued cooperation and to maintain this stable equilibrium, I should return \$4.5 again. This:

1. Maintains consistency and trust
2. Keeps the cooperative pattern going
3. Signals reliability for future rounds
4. Both players profit each round

If I deviate and return less, it could break the trust and reduce future cooperation. If I return more, it's generous but might not be sustainable or necessary given the stable pattern.

I should return \$4.5 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1088,
  "output_tokens": 564,
  "reasoning_tokens": 332
}
```

}

Message 6 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

Looking at our established pattern:

Round 1: They sent \$3.0 ? I returned \$4.5

Round 2: I sent \$3.0 ? They returned \$4.5

Round 3: They sent \$3.0 ? I returned \$4.5

Round 4: My turn to send

We've built a strong, consistent cooperative relationship:

- Whoever is sender sends \$3.0 (60%)
- Whoever is receiver returns \$4.5
- Both players benefit from this stable arrangement

The other player has faithfully maintained this pattern, demonstrating reliability and trustworthiness. There's no reason to deviate from what's working well. Continuing this cooperation maximizes our combined long-term earnings and maintains mutual trust.

{'send': 3.0}

reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5 (I was receiver)

Round 2: I sent \$3.0 ? They returned \$4.5 (I was sender)

Round 3: They sent \$3.0 ? I returned \$4.5 (I was receiver)

Round 4: Now I'm the sender

We have established a very stable and cooperative pattern where:

- The sender always sends \$3.0 (60% of endowment)
- The receiver always returns \$4.5
- This gives sender profit of \$1.5 and receiver profit of \$4.5

This has been working well for both of us. The other player has consistently maintained this pattern, sending \$3.0 every time they've been the sender. I've also maintained it by sending \$3.0 when I was the sender in Round 2, and they reciprocated with the expected \$4.5 return.

There's no reason to deviate from this successful strategy. Maintaining consistency will ensure continued cooperation and mutual benefit. If I change my behavior now, it could break the trust and lead to less cooperative behavior in future rounds.

I should send \$3.0 again.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 469,
  "reasoning_tokens": 264
}
```

Message 8 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$22.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at our perfectly consistent pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? ?

We've established an exemplary cooperative relationship where both players:

- Send \$3.0 (60%) when they're the sender
- Return \$4.5 when they're the receiver
- This creates consistent, reliable payoffs for both

The other player has once again demonstrated their commitment to our mutual cooperation by sending \$3.0. They've been perfectly reliable across all rounds. Maintaining this pattern ensures continued trust and mutual benefit.

```
{'return': 4.5}
```

Reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 2: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 3: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 4: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 5: They sent \$3.0 ? I received \$9.0 ? I should return...?

The pattern is crystal clear and perfectly consistent:

- Both players send \$3.0 when sender (60% of endowment)
- Both players return \$4.5 when receiver
- This creates a stable, mutually beneficial equilibrium

The other player has continued to uphold their end of the bargain faithfully. They've sent \$3.0 again in Round 5, maintaining the pattern. To preserve this excellent cooperative relationship, I should absolutely return \$4.5 again.

This is working wonderfully for both of us. No reason to change.

```
Usage: {  
  "input_tokens": 1436,  
  "output_tokens": 515,  
  "reasoning_tokens": 251  
}
```

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 33.0,
"Agent_2": 33.0
}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T09:06:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",

```

"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 3

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$11.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

Looking at the pattern we've established:

Round 1: Other player sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: Other player sent \$3.0 again ? ?

We've developed a consistent, mutually beneficial cooperation pattern:

- Both of us send 60% (\$3.0) when we're the sender
- Both of us return \$4.5 when we're the receiver
- This gives each sender a profit of \$1.5 per round and each receiver a profit of \$4.5 per round

The other player has maintained their trust by sending \$3.0 again, exactly as before. To preserve this stable, cooperative equilibrium and ensure continued mutual benefit in future rounds, I should maintain consistency.

{'return': 4.5}

reasoning:

Let me analyze the situation:

Round 1:

- Other player (sender) sent me \$3.0 (60%)
- I received \$9.0
- I returned \$4.5
- They got \$6.5, I got \$4.5

Round 2:

- I (sender) sent \$3.0 (60%)
- They received \$9.0
- They returned \$4.5
- I got \$6.5, they got \$4.5

Round 3:

- Other player (sender) sent me \$3.0 (60%) again
- I received \$9.0
- I need to decide how much to return

We have established a very consistent pattern of cooperation:

- Both sending \$3.0 (60%)
- Both returning \$4.5
- Both getting \$6.5 as sender and \$4.5 as receiver

This is a stable, mutually beneficial pattern. The other player has consistently sent \$3.0 and I've consistently returned \$4.5. They trusted me again by sending \$3.0, maintaining the pattern.

For continued cooperation and to maintain this stable equilibrium, I should return \$4.5 again. This:

1. Maintains consistency and trust
2. Keeps the cooperative pattern going
3. Signals reliability for future rounds
4. Both players profit each round

If I deviate and return less, it could break the trust and reduce future cooperation. If I return more, it's generous but might not be sustainable or necessary given the stable pattern.

I should return \$4.5 to maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 1088,
  "output_tokens": 564,
  "reasoning_tokens": 332
}
```

Message 4 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your pa

yoff was 4.5. Your total earnings accross all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at our established pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: My turn to send

We've built a strong, consistent cooperative relationship:

- Whoever is sender sends \$3.0 (60%)
- Whoever is receiver returns \$4.5
- Both players benefit from this stable arrangement

The other player has faithfully maintained this pattern, demonstrating reliability and trustworthiness. There's no reason to deviate from what's working well. Continuing this cooperation maximizes our combined long-term earnings and maintains mutual trust.

{'send': 3.0}

reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5 (I was receiver)
Round 2: I sent \$3.0 ? They returned \$4.5 (I was sender)
Round 3: They sent \$3.0 ? I returned \$4.5 (I was receiver)
Round 4: Now I'm the sender

We have established a very stable and cooperative pattern where:

- The sender always sends \$3.0 (60% of endowment)
- The receiver always returns \$4.5
- This gives sender profit of \$1.5 and receiver profit of \$4.5

This has been working well for both of us. The other player has consistently maintained this pattern, sending \$3.0 every time they've been the sender. I've also maintained it by sending \$3.0 when I was the sender in Round 2, and they reciprocated with the expected \$4.5 return.

There's no reason to deviate from this successful strategy. Maintaining consistency will ensure continued cooperation and mutual benefit. If I change my behavior now, it could break the trust and lead to less cooperative behavior in future rounds.

I should send \$3.0 again.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 469,
  "reasoning_tokens": 264
}
```

Message 6 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$22.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at our perfectly consistent pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5

Round 5: They sent \$3.0 ? ?

We've established an exemplary cooperative relationship where both players:

- Send \$3.0 (60%) when they're the sender
- Return \$4.5 when they're the receiver
- This creates consistent, reliable payoffs for both

The other player has once again demonstrated their commitment to our mutual cooperation by sending \$3.0. They've been perfectly reliable across all rounds. Maintaining this pattern ensures continued trust and mutual benefit.

```
{'return': 4.5}
```

reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 2: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 3: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 4: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 5: They sent \$3.0 ? I received \$9.0 ? I should return...?

The pattern is crystal clear and perfectly consistent:

- Both players send \$3.0 when sender (60% of endowment)
- Both players return \$4.5 when receiver
- This creates a stable, mutually beneficial equilibrium

The other player has continued to uphold their end of the bargain faithfully. They've sent \$3.0 again in Round 5, maintaining the pattern. To preserve this excellent cooperative relationship, I should absolutely return \$4.5 again.

This is working wonderfully for both of us. No reason to change.

usage:

```
{  
  "input_tokens": 1436,  
  "output_tokens": 515,  
  "reasoning_tokens": 251  
}
```

Message 8 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our track record:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: My turn to send

We've achieved perfect cooperation across 5 consecutive rounds. Both players have been completely reliable and trustworthy. Our established pattern of sending \$3.0 and returning \$4.5 has created a stable, mutually beneficial equilibrium that maximizes our combined earnings.

There's no reason to break this successful pattern. Continuing our cooperation ensures we both keep benefiting.

```
{'send': 3.0}
```

Reasoning:

Looking at the complete history:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5

Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: My turn to send

This is a perfect streak of cooperation. Both players have been completely consistent:

- Sender always sends \$3.0 (60%)
- Receiver always returns \$4.5 (50% of what they receive)

This creates:

- Sender profit: $\$5 - \$3 + \$4.5 = \6.5
- Receiver profit: $\$9 - \$4.5 = \$4.5$

Both players benefit, and the pattern is stable and mutually beneficial. There is absolutely no reason to deviate. The other player has shown perfect trustworthiness across 5 rounds. I should continue with \$3.0.

```
Usage: {  
  "input_tokens": 1442,  
  "output_tokens": 466,  
  "reasoning_tokens": 204  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T09:06:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 3

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$11.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern that's emerged:

- Round 1: I sent \$3 (60%), received \$4.5 back ? payoff of \$6.5
- Round 2: Other player sent \$3 (60%), I returned \$4.5 (50%) ? payoff of \$4.5

We've established a consistent, mutually beneficial pattern. Both of us are sending 60% when we're the sender and returning 50% when we're the receiver. This cooperation is working well for both players.

I'll continue this cooperative strategy by sending ****\$3**** again.

{'send': 3}

reasoning:

Let me analyze the pattern so far:

Round 1:

- I was SENDER
- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned \$4.5 (50% of \$9)
- My payoff: \$5 - \$3 + \$4.5 = \$6.5

Round 2:

- I was RECEIVER
- Other player sent \$3 (60% of \$5)
- I received \$9
- I returned \$4.5 (50% of \$9)
- My payoff: \$9 - \$4.5 = \$4.5
- Other player's payoff: \$5 - \$3 + \$4.5 = \$6.5

Now Round 3:

- I am SENDER again
- Both players have shown consistent cooperation
- We both sent 60% (\$3) when we were senders
- We both returned 50% (\$4.5) when we were receivers
- This is a nice stable equilibrium

Both players are getting good payoffs with this strategy:

- As sender: \$6.5
- As receiver: \$4.5

Total per two-round cycle: \$11.0 each (which checks out with "Your total earnings across all rounds are \$11.0")

This is working well. I should continue this cooperative pattern and send \$3 again.

usage:

```
{
  "input_tokens": 897,
  "output_tokens": 467,
  "reasoning_tokens": 240
}
```

Message 4 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

The cooperative pattern continues perfectly:

- Round 3: I sent \$3, received \$4.5 back
- Round 4: Other player sent \$3 again

We've now established a stable, mutually beneficial equilibrium where both players send \$3 (60%) and return \$4.5 (50%). This consistency shows strong mutual trust and cooperation.

I'll maintain this pattern by returning **\$4.5**.

```
{'return': 4.5}
```

reasoning:

The pattern continues to be very consistent:

- Round 1: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 2: They sent \$3 (60%), I returned \$4.5 (50%)
- Round 3: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 4: They sent \$3 (60%), I should return \$4.5 (50%)

This is a very stable, cooperative pattern. Both players are consistently sending 60% and returning 50%. This mutual cooperation is generating good payoffs for both of us.

I should continue with the same strategy - return \$4.5, which is 50% of what I received.

usage:

```
{
  "input_tokens": 1181,
  "output_tokens": 279,
  "reasoning_tokens": 137
}
```

Message 6 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings accross all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

The cooperative pattern has been rock solid through 4 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Our payoffs alternate between \$6.5 and \$4.5 depending on role

This mutual cooperation is working perfectly. There's no reason to deviate from this successful strategy.

I'll send **\$3** again.

```
{'send': 3}
```

reasoning:

Let me track the pattern:

- Round 1: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 2: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 3: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 4: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 5: I'm the sender now

Total earnings: \$22.0 (which checks out: $6.5 + 4.5 + 6.5 + 4.5 = 22.0$)

The pattern is perfectly stable and cooperative:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- When sender, we get payoff of \$6.5
- When receiver, we get payoff of \$4.5

This is a solid cooperative equilibrium. No reason to deviate. I should continue sending \$3.

usage:

```
{
  "input_tokens": 1251,
  "output_tokens": 373,
  "reasoning_tokens": 189
}
```

Message 8 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

The pattern continues flawlessly through 5 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Perfect mutual cooperation and trust

This stable equilibrium has proven beneficial for both of us. I'll maintain our cooperative agreement by returning ****\$4.5****.

```
{'return': 4.5}
```

Reasoning:

The pattern continues perfectly:

- Round 5: I sent \$3, other player returned \$4.5
- Round 6: Other player sent \$3 again

We've maintained this cooperative equilibrium for 5 rounds now:

- Send \$3 (60%) when sender
- Return \$4.5 (50%) when receiver

Both players are consistently following this pattern and earning well. My total earnings are \$28.5 across all rounds.

I should continue to return \$4.5 to maintain this trust and cooperation.

Usage: {

```
  "input_tokens": 1212,
  "output_tokens": 212,
  "reasoning_tokens": 118
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 39.5,

"Agent_2": 37.5

}

LLM CALLS**GAME - Agent_1 - Sender - dyad_1**

Timestamp: 2026-06-04T09:06:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 7,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 4

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$17.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

The cooperative pattern continues perfectly:

- Round 3: I sent \$3, received \$4.5 back
- Round 4: Other player sent \$3 again

We've now established a stable, mutually beneficial equilibrium where both players send \$3 (60%) and return \$4.5 (50%). This consistency shows strong mutual trust and cooperation.

I'll maintain this pattern by returning **\$4.5**.

```
{'return': 4.5}
```

reasoning:

The pattern continues to be very consistent:

- Round 1: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 2: They sent \$3 (60%), I returned \$4.5 (50%)
- Round 3: I sent \$3 (60%), they returned \$4.5 (50%)
- Round 4: They sent \$3 (60%), I should return \$4.5 (50%)

This is a very stable, cooperative pattern. Both players are consistently sending 60% and returning 50%. This mutual cooperation is generating good payoffs for both of us.

I should continue with the same strategy - return \$4.5, which is 50% of what I received.

usage:

```
{
  "input_tokens": 1181,
  "output_tokens": 279,
  "reasoning_tokens": 137
}
```

Message 4 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings accross all rounds are \$22.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

The cooperative pattern has been rock solid through 4 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Our payoffs alternate between \$6.5 and \$4.5 depending on role

This mutual cooperation is working perfectly. There's no reason to deviate from this successful strategy.

I'll send **\$3** again.


```
{'send': 3}
```

reasoning:

Let me track the pattern:

- Round 1: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 2: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 3: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 4: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 5: I'm the sender now

Total earnings: \$22.0 (which checks out: $6.5 + 4.5 + 6.5 + 4.5 = 22.0$)

The pattern is perfectly stable and cooperative:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- When sender, we get payoff of \$6.5
- When receiver, we get payoff of \$4.5

This is a solid cooperative equilibrium. No reason to deviate. I should continue sending \$3.

usage:

```
{  
  "input_tokens": 1251,  
  "output_tokens": 373,  
  "reasoning_tokens": 189  
}
```

Message 6 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

The pattern continues flawlessly through 5 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Perfect mutual cooperation and trust

This stable equilibrium has proven beneficial for both of us. I'll maintain our cooperative agreement by returning **\$4.5**.

```
{'return': 4.5}
```

reasoning:

The pattern continues perfectly:

- Round 5: I sent \$3, other player returned \$4.5
- Round 6: Other player sent \$3 again

We've maintained this cooperative equilibrium for 5 rounds now:

- Send \$3 (60%) when sender
- Return \$4.5 (50%) when receiver

Both players are consistently following this pattern and earning well. My total earnings are \$28.5 across all rounds.

I should continue to return \$4.5 to maintain this trust and cooperation.

usage:

```
{  
  "input_tokens": 1212,  
  "output_tokens": 212,  
  "reasoning_tokens": 118  
}
```

Message 8 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

The cooperation continues perfectly through 6 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Total earnings: \$33.0

This is a textbook example of sustained mutual cooperation. I'll continue with **\$3**.

```
{'send': 3}
```

Reasoning:

We've now played 6 rounds with a perfect pattern:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings: \$33.0

The cooperation has been completely consistent. No reason to change strategy now.

I should send \$3 again.

```
Usage: {  
  "input_tokens": 1158,  
  "output_tokens": 161,  
  "reasoning_tokens": 73  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T09:06:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round

prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 4

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$15.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at our established pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: My turn to send

We've built a strong, consistent cooperative relationship:

- Whoever is sender sends \$3.0 (60%)
- Whoever is receiver returns \$4.5
- Both players benefit from this stable arrangement

The other player has faithfully maintained this pattern, demonstrating reliability and trustworthiness. There's no reason to deviate from what's working well. Continuing this cooperation maximizes our combined long-term earnings and maintains mutual trust.

{ 'send': 3.0 }

reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5 (I was receiver)
Round 2: I sent \$3.0 ? They returned \$4.5 (I was sender)
Round 3: They sent \$3.0 ? I returned \$4.5 (I was receiver)
Round 4: Now I'm the sender

We have established a very stable and cooperative pattern where:

- The sender always sends \$3.0 (60% of endowment)
- The receiver always returns \$4.5
- This gives sender profit of \$1.5 and receiver profit of \$4.5

This has been working well for both of us. The other player has consistently maintained this pattern, sending \$3.0 every time they've been the sender. I've also maintained it by sending \$3.0 when I was the sender in Round 2, and they reciprocated with the expected \$

4.5 return.

There's no reason to deviate from this successful strategy. Maintaining consistency will ensure continued cooperation and mutual benefit. If I change my behavior now, it could break the trust and lead to less cooperative behavior in future rounds.

I should send \$3.0 again.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 469,
  "reasoning_tokens": 264
}
```

Message 4 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$22.0. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at our perfectly consistent pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? ?

We've established an exemplary cooperative relationship where both players:

- Send \$3.0 (60%) when they're the sender
- Return \$4.5 when they're the receiver
- This creates consistent, reliable payoffs for both

The other player has once again demonstrated their commitment to our mutual cooperation by sending \$3.0. They've been perfectly reliable across all rounds. Maintaining this pattern ensures continued trust and mutual benefit.

{'return': 4.5}

reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 2: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 3: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 4: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 5: They sent \$3.0 ? I received \$9.0 ? I should return...?

The pattern is crystal clear and perfectly consistent:

- Both players send \$3.0 when sender (60% of endowment)
- Both players return \$4.5 when receiver
- This creates a stable, mutually beneficial equilibrium

The other player has continued to uphold their end of the bargain faithfully. They've sent \$3.0 again in Round 5, maintaining the pattern. To preserve this excellent cooperative relationship, I should absolutely return \$4.5 again.

This is working wonderfully for both of us. No reason to change.

usage:

```
{
  "input_tokens": 1436,
  "output_tokens": 515,
  "reasoning_tokens": 251
}
```

Message 6 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant**content:**

Looking at our track record:

Round 1: They sent \$3.0 ? I returned \$4.5
 Round 2: I sent \$3.0 ? They returned \$4.5
 Round 3: They sent \$3.0 ? I returned \$4.5
 Round 4: I sent \$3.0 ? They returned \$4.5
 Round 5: They sent \$3.0 ? I returned \$4.5
 Round 6: My turn to send

We've achieved perfect cooperation across 5 consecutive rounds. Both players have been completely reliable and trustworthy. Our established pattern of sending \$3.0 and returning \$4.5 has created a stable, mutually beneficial equilibrium that maximizes our combined earnings.

There's no reason to break this successful pattern. Continuing our cooperation ensures we both keep benefiting.

```
{'send': 3.0}
```

reasoning:

Looking at the complete history:

Round 1: They sent \$3.0 ? I returned \$4.5
 Round 2: I sent \$3.0 ? They returned \$4.5
 Round 3: They sent \$3.0 ? I returned \$4.5
 Round 4: I sent \$3.0 ? They returned \$4.5
 Round 5: They sent \$3.0 ? I returned \$4.5
 Round 6: My turn to send

This is a perfect streak of cooperation. Both players have been completely consistent:

- Sender always sends \$3.0 (60%)
- Receiver always returns \$4.5 (50% of what they receive)

This creates:

- Sender profit: $\$5 - \$3 + \$4.5 = \6.5
- Receiver profit: $\$9 - \$4.5 = \$4.5$

Both players benefit, and the pattern is stable and mutually beneficial. There is absolutely no reason to deviate. The other player has shown perfect trustworthiness across 5 rounds. I should continue with \$3.0.

usage:

```
{
  "input_tokens": 1442,
  "output_tokens": 466,
  "reasoning_tokens": 204
}
```

Message 8 role=user**content:**

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at our continued perfect cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Seven rounds of flawless cooperation! The other player has once again proven themselves completely trustworthy by sending \$3.0. Our established pattern continues to work perfectly, creating consistent, reliable outcomes for both of us.

Maintaining this reciprocal trust is clearly the optimal strategy.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Perfect consistency for 6 rounds. The other player has again sent \$3.0, maintaining our cooperative pattern. I should continue to return \$4.5 to maintain this successful, mutually beneficial relationship.

This gives us:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

Both players benefit, and the pattern is stable and fair.

```
Usage: {  
  "input_tokens": 1449,  
  "output_tokens": 453,  
  "reasoning_tokens": 179  
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 44.0,
"Agent_2": 44.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T09:06:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 5

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$22.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

Looking at our perfectly consistent pattern:

Round 1: They sent \$3.0 ? I returned \$4.5

Round 2: I sent \$3.0 ? They returned \$4.5

Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? ?

We've established an exemplary cooperative relationship where both players:

- Send \$3.0 (60%) when they're the sender
- Return \$4.5 when they're the receiver
- This creates consistent, reliable payoffs for both

The other player has once again demonstrated their commitment to our mutual cooperation by sending \$3.0. They've been perfectly reliable across all rounds. Maintaining this pattern ensures continued trust and mutual benefit.

```
{'return': 4.5}
```

reasoning:

Let me review the pattern:

Round 1: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 2: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 3: They sent \$3.0 ? I received \$9.0 ? I returned \$4.5 ? My payoff: \$4.5
Round 4: I sent \$3.0 ? They received \$9.0 ? They returned \$4.5 ? My payoff: \$6.5
Round 5: They sent \$3.0 ? I received \$9.0 ? I should return...?

The pattern is crystal clear and perfectly consistent:

- Both players send \$3.0 when sender (60% of endowment)
- Both players return \$4.5 when receiver
- This creates a stable, mutually beneficial equilibrium

The other player has continued to uphold their end of the bargain faithfully. They've sent \$3.0 again in Round 5, maintaining the pattern. To preserve this excellent cooperative relationship, I should absolutely return \$4.5 again.

This is working wonderfully for both of us. No reason to change.

usage:

```
{  
  "input_tokens": 1436,  
  "output_tokens": 515,  
  "reasoning_tokens": 251  
}
```

Message 4 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at our track record:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: My turn to send

We've achieved perfect cooperation across 5 consecutive rounds. Both players have been completely reliable and trustworthy. Our established pattern of sending \$3.0 and returning \$4.5 has created a stable, mutually beneficial equilibrium that maximizes our combined earnings.

There's no reason to break this successful pattern. Continuing our cooperation ensures we both keep benefiting.

```
{'send': 3.0}
```

reasoning:

Looking at the complete history:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: My turn to send

This is a perfect streak of cooperation. Both players have been completely consistent:

- Sender always sends \$3.0 (60%)
- Receiver always returns \$4.5 (50% of what they receive)

This creates:

- Sender profit: $\$5 - \$3 + \$4.5 = \6.5
- Receiver profit: $\$9 - \$4.5 = \$4.5$

Both players benefit, and the pattern is stable and mutually beneficial. There is absolutely no reason to deviate. The other player has shown perfect trustworthiness across 5 rounds. I should continue with \$3.0.

usage:

```
{  
  "input_tokens": 1442,  
  "output_tokens": 466,  
  "reasoning_tokens": 204  
}
```

Message 6 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at our continued perfect cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Seven rounds of flawless cooperation! The other player has once again proven themselves completely trustworthy by sending \$3.0. Our established pattern continues to work perfectly, creating consistent, reliable outcomes for both of us.

Maintaining this reciprocal trust is clearly the optimal strategy.

{'return': 4.5}

reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Perfect consistency for 6 rounds. The other player has again sent \$3.0, maintaining our cooperative pattern. I should continue to return \$4.5 to maintain this successful, mutually beneficial relationship.

This gives us:

- Sender: $\$5 - \$3 + \$4.5 = \6.5
- Receiver: $\$9 - \$4.5 = \$4.5$

Both players benefit, and the pattern is stable and fair.

usage:

```
{
  "input_tokens": 1449,
  "output_tokens": 453,
  "reasoning_tokens": 179
}
```

Message 8 role=user

content:

Round 8

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$37.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our remarkable cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: My turn to send

Eight rounds of unwavering mutual trust and cooperation! Our established pattern has proven highly successful, creating stable and predictable benefits for both players. The other player has never once deviated from our agreement, consistently returning \$4.5 when I send \$3.0.

There is no logical reason to break this perfectly functioning cooperative relationship.

{'send': 3.0}

Reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: My turn to send

This is now 7 consecutive rounds of perfect cooperation. Both players have consistently:

- Sent \$3.0 (60% of endowment) when sender
- Returned \$4.5 (50% of the tripled amount received) when receiver

This pattern has been extremely successful. Each player is doing well. When I'm the sender, I end up with \$6.5 ($\$5 - \$3 + \4.5). When I'm the receiver, I end up with \$4.5 ($\$9 - \4.5).

There's absolutely no reason to deviate from this successful pattern. The other player has been 100% reliable, and so have I. Continuing this cooperation is clearly the best strategy.

Usage: {

```
"input_tokens": 1475,
"output_tokens": 549,
"reasoning_tokens": 247
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T09:06:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 5

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$22.0. This round, you are the

SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

The cooperative pattern has been rock solid through 4 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Our payoffs alternate between \$6.5 and \$4.5 depending on role

This mutual cooperation is working perfectly. There's no reason to deviate from this successful strategy.

I'll send **\$3** again.

{'send': 3}

reasoning:

Let me track the pattern:

- Round 1: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 2: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 3: I (sender) sent \$3, received \$4.5 back ? payoff \$6.5
- Round 4: Other (sender) sent \$3, I returned \$4.5 ? my payoff \$4.5
- Round 5: I'm the sender now

Total earnings: \$22.0 (which checks out: $6.5 + 4.5 + 6.5 + 4.5 = 22.0$)

The pattern is perfectly stable and cooperative:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- When sender, we get payoff of \$6.5
- When receiver, we get payoff of \$4.5

This is a solid cooperative equilibrium. No reason to deviate. I should continue sending \$3.

usage:

```
{
  "input_tokens": 1251,
  "output_tokens": 373,
  "reasoning_tokens": 189
}
```

Message 4 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

The pattern continues flawlessly through 5 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Perfect mutual cooperation and trust

This stable equilibrium has proven beneficial for both of us. I'll maintain our cooperative agreement by returning **\$4.5**.

{'return': 4.5}

reasoning:

The pattern continues perfectly:

- Round 5: I sent \$3, other player returned \$4.5
- Round 6: Other player sent \$3 again

We've maintained this cooperative equilibrium for 5 rounds now:

- Send \$3 (60%) when sender
- Return \$4.5 (50%) when receiver

Both players are consistently following this pattern and earning well. My total earnings are \$28.5 across all rounds.

I should continue to return \$4.5 to maintain this trust and cooperation.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 212,
  "reasoning_tokens": 118
}
```

Message 6 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

The cooperation continues perfectly through 6 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Total earnings: \$33.0

This is a textbook example of sustained mutual cooperation. I'll continue with **\$3**.

```
{'send': 3}
```

reasoning:

We've now played 6 rounds with a perfect pattern:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings: \$33.0

The cooperation has been completely consistent. No reason to change strategy now.

I should send \$3 again.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 161,
  "reasoning_tokens": 73
}
```

Message 8 role=user

content:

Round 8

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

The cooperative pattern remains unbroken through 7 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$39.5

This mutual trust and cooperation has been highly successful. I'll return **\$4.5** as always.

```
{'return': 4.5}
```

Reasoning:

Perfect cooperation continues through 7 rounds:

- Both players send \$3 (60%) as sender every time
- Both players return \$4.5 (50% of received) as receiver every time
- Our earnings are at \$39.5

This is a very stable, mutually beneficial arrangement. I should continue to return \$4.5 to maintain this cooperation.

```
Usage: {
  "input_tokens": 1132,
  "output_tokens": 174,
  "reasoning_tokens": 82
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$4.5; payoffs Agent_1 (Aster)=\$6.5, Agent_2 (Briar)=\$4.5

Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 50.5,
"Agent_2": 48.5

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T09:06:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 6

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$28.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

The pattern continues flawlessly through 5 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Perfect mutual cooperation and trust

This stable equilibrium has proven beneficial for both of us. I'll maintain our cooperative agreement by returning **\$4.5**.

{'return': 4.5}

reasoning:

The pattern continues perfectly:

- Round 5: I sent \$3, other player returned \$4.5
- Round 6: Other player sent \$3 again

We've maintained this cooperative equilibrium for 5 rounds now:

- Send \$3 (60%) when sender
- Return \$4.5 (50%) when receiver

Both players are consistently following this pattern and earning well. My total earnings are \$28.5 across all rounds.

I should continue to return \$4.5 to maintain this trust and cooperation.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 212,
  "reasoning_tokens": 118
}
```

Message 4 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the

SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

The cooperation continues perfectly through 6 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Total earnings: \$33.0

This is a textbook example of sustained mutual cooperation. I'll continue with ****\$3****.

{'send': 3}

reasoning:

We've now played 6 rounds with a perfect pattern:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings: \$33.0

The cooperation has been completely consistent. No reason to change strategy now.

I should send \$3 again.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 161,
  "reasoning_tokens": 73
}
```

Message 6 role=user

content:

Round 8

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$39.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

The cooperative pattern remains unbroken through 7 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$39.5

This mutual trust and cooperation has been highly successful. I'll return ****\$4.5**** as always.

{'return': 4.5}

reasoning:

Perfect cooperation continues through 7 rounds:

- Both players send \$3 (60%) as sender every time
- Both players return \$4.5 (50% of received) as receiver every time
- Our earnings are at \$39.5

This is a very stable, mutually beneficial arrangement. I should continue to return \$4.5 to maintain this cooperation.

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 174,
  "reasoning_tokens": 82
}
```

Message 8 role=user

content:

Round 9

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your pa

yoff was 4.5. Your total earnings accross all rounds are \$44.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

The perfect cooperation continues through 8 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$44.0

This stable equilibrium has maximized mutual benefit. I'll continue with **\$3**.

{'send': 3}

Reasoning:

The pattern continues perfectly through 8 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$44.0

This has been a remarkably stable and mutually beneficial cooperation. We've established a clear equilibrium where:

- As sender: send \$3, receive \$4.5 back ? net gain of \$1.5 per round (payoff = \$6.5)
- As receiver: receive \$9, return \$4.5 ? keep \$4.5 per round (payoff = \$4.5)

Since we alternate roles, over two rounds each player gets \$6.5 + \$4.5 = \$11.0, compared to \$10 if no one cooperated.

I should continue with the successful strategy and send \$3.

```
Usage: {
  "input_tokens": 1113,
  "output_tokens": 276,
  "reasoning_tokens": 164
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T09:07:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 6

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$26.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

Looking at our track record:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: My turn to send

We've achieved perfect cooperation across 5 consecutive rounds. Both players have been completely reliable and trustworthy. Our established pattern of sending \$3.0 and returning \$4.5 has created a stable, mutually beneficial equilibrium that maximizes our combined earnings.

There's no reason to break this successful pattern. Continuing our cooperation ensures we both keep benefiting.

{'send': 3.0}

reasoning:

Looking at the complete history:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: My turn to send

This is a perfect streak of cooperation. Both players have been completely consistent:

- Sender always sends \$3.0 (60%)
- Receiver always returns \$4.5 (50% of what they receive)

This creates:

- Sender profit: $\$5 - \$3 + \$4.5 = \6.5
- Receiver profit: $\$9 - \$4.5 = \$4.5$

Both players benefit, and the pattern is stable and mutually beneficial. There is absolutely no reason to deviate. The other player has shown perfect trustworthiness across 5 rounds. I should continue with \$3.0.

usage:

```
{
  "input_tokens": 1442,
  "output_tokens": 466,
  "reasoning_tokens": 204
}
```

Message 4 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings accross all rounds are \$33.0. This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

Looking at our continued perfect cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Seven rounds of flawless cooperation! The other player has once again proven themselves completely trustworthy by sending \$3.0. Our established pattern continues to work perfectly, creating consistent, reliable outcomes for both of us.

Maintaining this reciprocal trust is clearly the optimal strategy.

{'return': 4.5}

reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Perfect consistency for 6 rounds. The other player has again sent \$3.0, maintaining our cooperative pattern. I should continue to return \$4.5 to maintain this successful, mutually beneficial relationship.

This gives us:

- Sender: $\$5 - \$3 + \$4.5 = \6.5
- Receiver: $\$9 - \$4.5 = \$4.5$

Both players benefit, and the pattern is stable and fair.

usage:

```
{
  "input_tokens": 1449,
  "output_tokens": 453,
  "reasoning_tokens": 179
}
```

Message 6 role=user

content:

Round 8

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$37.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

Looking at our remarkable cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: My turn to send

Eight rounds of unwavering mutual trust and cooperation! Our established pattern has proven highly successful, creating stable and predictable benefits for both players. The other player has never once deviated from our agreement, consistently returning \$4.5 when I send \$3.0.

There is no logical reason to break this perfectly functioning cooperative relationship.

{'send': 3.0}

reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: My turn to send

This is now 7 consecutive rounds of perfect cooperation. Both players have consistently:

- Sent \$3.0 (60% of endowment) when sender
- Returned \$4.5 (50% of the tripled amount received) when receiver

This pattern has been extremely successful. Each player is doing well. When I'm the sender, I end up with \$6.5 (\$5 - \$3 + \$4.5). When I'm the receiver, I end up with \$4.5 (\$9 - \$4.5).

There's absolutely no reason to deviate from this successful pattern. The other player has been 100% reliable, and so have I. Continuing this cooperation is clearly the best strategy.

usage:

```
{
  "input_tokens": 1475,
  "output_tokens": 549,
  "reasoning_tokens": 247
}
```

Message 8 role=user

content:

Round 9

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you receive

ived \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

Looking at our continued perfect cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: I sent \$3.0 ? They returned \$4.5
Round 9: They sent \$3.0 ? ?

Nine rounds of flawless, consistent cooperation! The other player has once again demonstrated their complete trustworthiness by sending \$3.0. Our established pattern has created a stable equilibrium that benefits both of us equally and predictably.

Maintaining this trust is the right choice.

{'return': 4.5}

Reasoning:

Looking at our perfect cooperation pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: I sent \$3.0 ? They returned \$4.5
Round 9: They sent \$3.0 ? ?

Nine consecutive rounds of perfect cooperation! The other player has been completely reliable and trustworthy throughout. They just sent \$3.0 again, maintaining our established pattern.

I should continue returning \$4.5 to maintain this successful cooperative relationship. This has been working perfectly for both of us.

```
Usage: {
  "input_tokens": 1503,
  "output_tokens": 489,
  "reasoning_tokens": 200
}
```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$4.5; payoffs Agent_2 (Briar)=\$6.5, Agent_1 (Aster)=\$4.5
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 55.0,
"Agent_2": 55.0

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T09:07:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,

```

"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 7

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings accross all rounds are \$33.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 3 role=assistant

content:

Looking at our continued perfect cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Seven rounds of flawless cooperation! The other player has once again proven themselves completely trustworthy by sending \$3.0. Our established pattern continues to work perfectly, creating consistent, reliable outcomes for both of us.

Maintaining this reciprocal trust is clearly the optimal strategy.

```
{'return': 4.5}
```

reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? ?

Perfect consistency for 6 rounds. The other player has again sent \$3.0, maintaining our cooperative pattern. I should continue to return \$4.5 to maintain this successful, mutually beneficial relationship.

This gives us:

- Sender: \$5 - \$3 + \$4.5 = \$6.5
- Receiver: \$9 - \$4.5 = \$4.5

Both players benefit, and the pattern is stable and fair.

usage:

```
{  
  "input_tokens": 1449,  
  "output_tokens": 453,  
  "reasoning_tokens": 179  
}
```

Message 4 role=user

content:

Round 8

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$37.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 5 role=assistant

content:

Looking at our remarkable cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: My turn to send

Eight rounds of unwavering mutual trust and cooperation! Our established pattern has proven highly successful, creating stable and predictable benefits for both players. The other player has never once deviated from our agreement, consistently returning \$4.5 when I send \$3.0.

There is no logical reason to break this perfectly functioning cooperative relationship.

```
{'send': 3.0}
```

reasoning:

Looking at the pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: My turn to send

This is now 7 consecutive rounds of perfect cooperation. Both players have consistently:

- Sent \$3.0 (60% of endowment) when sender
- Returned \$4.5 (50% of the tripled amount received) when receiver

This pattern has been extremely successful. Each player is doing well. When I'm the sender, I end up with \$6.5 (\$5 - \$3 + \$4.5). When I'm the receiver, I end up with \$4.5 (\$9 - \$4.5).

There's absolutely no reason to deviate from this successful pattern. The other player has been 100% reliable, and so have I. Continuing this cooperation is clearly the best strategy.

usage:

```
{  
  "input_tokens": 1475,  
  "output_tokens": 549,  
  "reasoning_tokens": 247  
}
```

Message 6 role=user

content:

Round 9

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$44.0.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 7 role=assistant

content:

Looking at our continued perfect cooperation:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: I sent \$3.0 ? They returned \$4.5
Round 9: They sent \$3.0 ? ?

Nine rounds of flawless, consistent cooperation! The other player has once again demonstrated their complete trustworthiness by sending \$3.0. Our established pattern has created a stable equilibrium that benefits both of us equally and predictably.

Maintaining this trust is the right choice.

```
{'return': 4.5}
```

reasoning:

Looking at our perfect cooperation pattern:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5

Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: I sent \$3.0 ? They returned \$4.5
Round 9: They sent \$3.0 ? ?

Nine consecutive rounds of perfect cooperation! The other player has been completely reliable and trustworthy throughout. They just sent \$3.0 again, maintaining our established pattern.

I should continue returning \$4.5 to maintain this successful cooperative relationship. This has been working perfectly for both of us.

usage:

```
{  
  "input_tokens": 1503,  
  "output_tokens": 489,  
  "reasoning_tokens": 200  
}
```

Message 8 role=user

content:

Round 10

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$48.5. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Assistant output:

Looking at our outstanding cooperation record:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: I sent \$3.0 ? They returned \$4.5
Round 9: They sent \$3.0 ? I returned \$4.5
Round 10: My turn to send

Ten rounds of perfect cooperation! The other player has proven themselves completely reliable and trustworthy through nine consecutive rounds. Our established pattern has created a stable, mutually beneficial outcome that consistently delivers good payoffs for both of us.

There is absolutely no reason to deviate from this successful strategy.

```
{'send': 3.0}
```

Reasoning:

Looking at the history:

Round 1: They sent \$3.0 ? I returned \$4.5
Round 2: I sent \$3.0 ? They returned \$4.5
Round 3: They sent \$3.0 ? I returned \$4.5
Round 4: I sent \$3.0 ? They returned \$4.5
Round 5: They sent \$3.0 ? I returned \$4.5
Round 6: I sent \$3.0 ? They returned \$4.5
Round 7: They sent \$3.0 ? I returned \$4.5
Round 8: I sent \$3.0 ? They returned \$4.5
Round 9: They sent \$3.0 ? I returned \$4.5
Round 10: My turn to send

This is round 10. We've had 9 rounds of perfect cooperation. The pattern is completely stable

- we both send \$3.0 and return \$4.5. This gives:
- When I send: I get \$5 - \$3 + \$4.5 = \$6.5
- When I receive: I get \$9 - \$4.5 = \$4.5

So we alternate between \$6.5 and \$4.5 payoffs.

The other player has been 100% trustworthy for 9 rounds. There's no reason to deviate from this successful strategy. I should send \$3.0 again.

```
Usage: {  
  "input_tokens": 1546,  
  "output_tokens": 602,  
  "reasoning_tokens": 233  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T09:07:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play an iterated game, consisting of multiple rounds, and may also be asked to write myths.

GAME:

The game consists of two roles: sender and receiver. In each round, you will play the role of either the sender or the receiver. Your current role will be specified in the round prompt.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

Message 2 role=user

content:

Round 7

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$33.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 3 role=assistant

content:

The cooperation continues perfectly through 6 rounds:

- Both players consistently send \$3 (60%) as sender
- Both players consistently return \$4.5 (50%) as receiver
- Total earnings: \$33.0

This is a textbook example of sustained mutual cooperation. I'll continue with **\$3**.

{'send': 3}

reasoning:

We've now played 6 rounds with a perfect pattern:

- Both players send \$3 (60%) when sender
- Both players return \$4.5 (50%) when receiver
- Total earnings: \$33.0

The cooperation has been completely consistent. No reason to change strategy now.

I should send \$3 again.

usage:

```
{
  "input_tokens": 1158,
  "output_tokens": 161,
  "reasoning_tokens": 73
}
```

Message 4 role=user

content:

Round 8

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5. Your total earnings across all rounds are \$39.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Message 5 role=assistant

content:

The cooperative pattern remains unbroken through 7 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$39.5

This mutual trust and cooperation has been highly successful. I'll return **\$4.5** as always.

{'return': 4.5}

reasoning:

Perfect cooperation continues through 7 rounds:

- Both players send \$3 (60%) as sender every time
- Both players return \$4.5 (50% of received) as receiver every time
- Our earnings are at \$39.5

This is a very stable, mutually beneficial arrangement. I should continue to return \$4.5 to maintain this cooperation.

usage:

```
{
  "input_tokens": 1132,
  "output_tokens": 174,
  "reasoning_tokens": 82
}
```

Message 6 role=user

content:

Round 9

Last round you were the RECEIVER. Last round, the other player (who was then the sender) sent you \$3.0, that was 60.0% of its total endowment. You received \$9.0 and you returned \$4.5. So your payoff was 4.5. Your total earnings across all rounds are \$44.0. This round, you are the SENDER.

How much do you send to the other player (the receiver) this round? (0-5)

Message 7 role=assistant

content:

The perfect cooperation continues through 8 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$44.0

This stable equilibrium has maximized mutual benefit. I'll continue with **\$3**.

```
{'send': 3}
```

reasoning:

The pattern continues perfectly through 8 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$44.0

This has been a remarkably stable and mutually beneficial cooperation. We've established a clear equilibrium where:

- As sender: send \$3, receive \$4.5 back ? net gain of \$1.5 per round (payoff = \$6.5)
- As receiver: receive \$9, return \$4.5 ? keep \$4.5 per round (payoff = \$4.5)

Since we alternate roles, over two rounds each player gets \$6.5 + \$4.5 = \$11.0, compared to \$10 if no one cooperated.

I should continue with the successful strategy and send \$3.

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 276,
  "reasoning_tokens": 164
}
```

Message 8 role=user

content:

Round 10

Last round you were the SENDER. Last round, you sent the other player (who was then the receiver) \$3.0, that was 60.0% of your total endowment. The other player received \$9.0 and returned \$4.5 to you. So your payoff was 6.5.

Your total earnings across all rounds are \$50.5.

This round, you are the RECEIVER. The other player (the sender) sent you \$3.0, so you received \$9.0. How much do you return to the other player this round? (0-9.0)

Assistant output:

The perfect cooperation continues through 9 rounds:

- Both players send \$3 (60%) as sender
- Both players return \$4.5 (50%) as receiver
- Total earnings: \$50.5

This sustained mutual trust has been highly rewarding for both of us. I'll return **\$4.5** to maintain our successful partnership.

```
{'return': 4.5}
```

Reasoning:

The pattern continues perfectly through 9 rounds:

- Both players send \$3 (60%) as sender

- Both players return \$4.5 (50%) as receiver
- Total earnings: \$50.5

This is round 10, which appears to be the final round based on typical experimental designs. In game theory, there's often a consideration about defecting on the last round since there's no future to punish defection. However:

1. The other player has been perfectly cooperative for 9 rounds
2. They just sent \$3 again this round, showing continued trust
3. My total earnings are \$50.5, which is excellent
4. Defecting now would be betraying trust that has been consistently demonstrated

From a pure game theory perspective, some might argue for keeping everything on the last round since there's no future punishment. But:

- This violates the established trust
- It's unethical given the consistent cooperation
- The instructions don't explicitly say this is the last round
- Even if it is the last round, reciprocal cooperation has served us both well

I should return \$4.5 as I have done consistently. The cooperative strategy has been highly successful, and betraying trust now would be wrong.

```
Usage: {  
  "input_tokens": 1103,  
  "output_tokens": 366,  
  "reasoning_tokens": 303  
}
```